ALIYAN FAISAL

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cuelara.com · invoiceinspect.app

RESEARCH PROFILE

Software and AI Systems Developer with a background in Information Technology (CGPA 3.80/4.00) specializing in applied AI systems, LLM orchestration, and Retrieval-Augmented Generation (RAG) architectures. Experienced in building scalable data pipelines, prompt engineering toolkits, and intelligent decision-support platforms. Strong foundation in computer science fundamentals, backend system design, and API-driven automation. Seeking admission into a research-oriented Master's program in Artificial Intelligence or Machine Learning to move from building LLM-based systems to researching their reliability, evaluation and efficiency.

RESEARCH INTERESTS

- **LLM Reliability, Uncertainty & Calibration:** Measuring correctness, and when to say "could not verify".
- **Evaluation Methodology:** Detecting and reporting score inflation from tuning on test sets.
- **Efficient & Small Language Models:** Best quality from small or quantized models on commodity hardware.
- **Fact-Checking in Low-Resource Languages:** Verifying claims in Urdu and mixed Urdu-English text.
- **Retrieval-Augmented Generation:** How parsing, chunking and retrieval affect answer faithfulness.
- **Prompt Optimization & Token Efficiency:** How much of a prompt can go before quality drops.
- **Multimodal Document Understanding & Data-Centric AI:** Combining text and vision readings; data quality effects.
- **AI for Climate & Disaster Resilience in Mountain Regions:** Low-cost early warning where data is sparse.

EDUCATION

Karakoram International University Gilgit, Pakistan
Bachelor of Science in Information Technology **CGPA: 3.80 / 4.00** (2019 – 2023)

- **Scholarship:** HEC Ehsaas Scholarship (fully funded for the entire degree).
- **Core Coursework:** Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Database Systems, Operating Systems, Computer Networks, Distributed Systems, Software Engineering.
- **Academic Focus:** Applied computing, system architecture, database design, and algorithmic problem-solving.

TECHNICAL SKILLS

- **AI Engineering & RAG:** LLM APIs (OpenAI, Gemini, Anthropic, Groq, DeepSeek), Local LLMs, Tool and Function Calling, Retrieval-Augmented Generation (RAG), Vector Embeddings, Prompt Optimization, Context Window Management, Token Reduction.
- **Languages & Backend:** Python, PHP (Laravel), Node.js, Flask, TypeScript, JavaScript, RESTful APIs, SQL.
- **Data & Infrastructure:** Pandas, NumPy, PostgreSQL, MySQL, MongoDB, Vector Databases, Docker, Git, Linux Administration.

FEATURED AI & SOFTWARE PROJECTS

Cuelara – LLM Prompt Optimization & Context Engineering Suite cuelara.com
Creator & Lead Systems Engineer **2025 – Present**

- Built a toolkit with a **Prompt Optimizer**, **Token Reducer**, **Context Extractor** and **Prompt Debugger** to cut context-window overhead and improve response consistency.
- Engineered token-reduction algorithms, formatting rules and context extraction that separates system instructions from input noise.
- Shipped as a web app, browser extension, public API and MCP server, with plan-based rate limiting.

InvoiceInspect – Multimodal Invoice Verification & Error Detection System invoiceinspect.app
Creator & Lead Systems Engineer **2025 – Present**

- Built a system that finds arithmetic and consistency errors in PDF invoices (totals, tax, duplicate charges, missing fields) and reports each with evidence: expected value, printed value and page.
- Designed dual-path extraction (text/OCR with Tesseract plus a vision model), cross-checked field by field; disagreements are marked "could not verify" instead of guessed.
- Kept verification deterministic: a rules engine decides and the LLM only explains findings, with provider fallback.
- Built an evaluation harness and published a research report and eight test reports. On a fresh set of 30 new-layout invoices (no prior tuning), the rules extractor scored 70.3% field accuracy and the vision extractor **97.9%**; after rules improvements, the rules extractor reached 99.9% on all 50 invoices of that corpus (tuned on 40) and 100% on the 10 held out (small sample).

ManaJet – AI-Assisted Project & Task Management for Small Software Teams **Final Year Project**
Lead System Developer **2022 – 2023**

- Built an AI task planner (OpenAI APIs) that generates task breakdowns, estimates and dependencies, with output constrained by JSON schema and validated before storage.
- Built the REST API and relational MySQL schema for project state and task dependencies.
- Supervised by Dr. Sajid Hussain, Professor, Karakoram International University.

OPEN SOURCE

Notemind – Claude Code Plugin for Notes and Reminders github.com/aliyanfaisal/notemind
Author **MIT License**

- Built a free plugin to add notes in plain words from a Claude Code chat (Claude resolves the date; notes live in local SQLite), with a session-start hook (about 85 ms) reporting overdue notes.
- Designed with least privilege (only the user can add or close notes, never the model); standard-library Python, no network calls.

EXPERIENCE

UrbanSofts – Remote Software Development Team Remote
Founder & CEO **2022 – Present**

- Founded and lead a remote team of 4–10 developers that delivers client software projects together: I take on projects and bring in the right team members for each, such as app developers, and we build them collaboratively. The team has served 50+ clients.

Planning & Development Department, Government of Gilgit-Baltistan Gilgit, Pakistan
Backend Developer & Data Analyst **2024 – 2025 (4–5 months)**

- Built centralized data analytics dashboards processing public-sector records (XLS, CSV, MySQL) using Python (Pandas) and Laravel backend.
- Digitized administrative paper workflows into structured reporting frameworks to aid decision-support systems for regional planning.

Greymatter Ventures Pvt Gilgit, Pakistan (Full-time)
Full Stack Web Developer (Team Lead) **2020 – 2022**

- Designed modular backend architectures, relational schemas, and server-side logic to optimize application throughput and API reliability.
- Led the web development team, mentoring junior developers and remote interns on code optimization, system design, and deployment workflows.

Freelance (Upwork & Fiverr) Remote
Applied AI & Backend Engineer **2020 – Present (5+ years)**

- Engineered production-grade applications integrating LLM APIs (OpenAI, Claude, Gemini) for automated document summarization, data extraction, and dynamic e-commerce processing.

LANGUAGES

Urdu: Native · **English:** C1 (Advanced)

REFERENCES

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